Practical 1: Study of Different Dataset Types, Importing Using Python and Data Preprocessing Techniques

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Study of Different Dataset Types:

In [2]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')
```

In [3]:
```
df = pd.read_csv('gate_applicants.csv')
df
```

Out[3]:

	Unnamed: 0	Paper Code	Paper Name	Applied	Appeared	Year
0	0	AE	Aerospace Engineering	4639	3697	2019
1	1	AG	Agricultural Engineering	2088	1848	2019
2	2	AR	Architecture and Planning	11208	9105	2019
3	3	BT	Biotechnology	11295	9348	2019
4	4	CE	Civil Engineering	170999	145064	2019
...	...	...	...	...	...	...
65	65	PH	Physics	12243	9910	2016
66	66	PI	Production and Industrial Engineering	4131	3378	2016
67	67	TF	Textile Engineering and Fibre Science	1587	1406	2016
68	68	XE	Engineering Sciences	3722	3074	2016
69	69	XL	Life Sciences	12791	10298	2016

70 rows × 6 columns

In [4]:
```
df_excel = pd.read_excel("AB_NYC_2019.xlsx")

df_excel
```

Out[4]:

	id	name	host_id	host_name	neighbourhood_group	ne
0	2539	Clean & quiet apt home by the park	2787	John	Brooklyn	
1	2595	Skylit Midtown Castle	2845	Jennifer	Manhattan	
2	3647	THE VILLAGE OF HARLEM....NEW YORK !	4632	Elisabeth	Manhattan	
3	3831	Cozy Entire Floor of Brownstone	4869	LisaRoxanne	Brooklyn	
4	5022	Entire Apt: Spacious Studio/Loft by central park	7192	Laura	Manhattan	
...	...	...	...	...	...	
48890	36484665	Charming one bedroom - newly renovated rowhouse	8232441	Sabrina	Brooklyn	
48891	36485057	Affordable room in Bushwick/East Williamsburg	6570630	Marisol	Brooklyn	
48892	36485431	Sunny Studio at Historical Neighborhood	23492952	Ilgar & Aysel	Manhattan	
48893	36485609	43rd St. Time Square-cozy single bed	30985759	Taz	Manhattan	
48894	36487245	Trendy duplex in the very heart of Hell's Kitchen	68119814	Christophe	Manhattan	

48895 rows × 16 columns

In [5]:
```
df_json = pd.read_json("spam_dataset.json")
df_json
```

Out[5]:

	message_content	is_spam
0	Hello Lonnie,\n\nJust wanted to touch base reg...	0
1	Congratulations, you've won a prize! Call us n...	1
2	You have been pre-approved for a credit card w...	1
3	Limited time offer, act now! Only a few spots ...	1
4	Your loan has been approved! Transfer funds to...	1
...	...	...
995	Hello Virginia,\n\nIt was great to catch up wi...	0
996	Final notice: Claim your inheritance from a di...	1
997	Hot singles in your area want to chat! Contact...	1
998	Your loan has been approved! Transfer funds to...	1
999	Dear Brian,\n\nI wanted to let you know about ...	0

1000 rows × 2 columns

In [6]:
```
df_text = pd.read_csv("dataset.txt",delimiter="\t")

print(df_text.head())
```

```
   | Student_ID | Name   | Age | Marks |
0  | ---------- | -----  | --: | ----: |
1  | 101        | Milind   | 20 |  92   |
2  | 102        | Swayam |  21 |   94 |
3  | 103        | Shantanu| 21 |   92 |
4  | 104        | Atharv |  20 |   91 |
```

In [7]:
```
import cv2

image = cv2.imread("cat.jpg")

print(image.shape)
```

```
(720, 1280, 3)
```

Importing Using Python and Data Preprocessing Techniques:

In [10]:
```
df3 = pd.read_csv('salary_Data.csv')
df3
```

Out[10]:

	YearsExperience	Salary
0	1.1	39343.0
1	1.3	46205.0
2	1.5	37731.0
3	2.0	43525.0
4	2.2	39891.0
5	2.9	56642.0
6	3.0	60150.0
7	3.2	54445.0
8	3.2	64445.0
9	3.7	57189.0
10	3.9	63218.0
11	4.0	55794.0
12	4.0	56957.0
13	4.1	57081.0
14	4.5	61111.0
15	4.9	67938.0
16	5.1	66029.0
17	5.3	83088.0
18	5.9	81363.0
19	6.0	93940.0
20	6.8	91738.0
21	7.1	98273.0
22	7.9	101302.0
23	8.2	113812.0
24	8.7	109431.0
25	9.0	105582.0
26	9.5	116969.0
27	9.6	112635.0
28	10.3	122391.0
29	10.5	121872.0

In [11]:
```
df3.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     float64
dtypes: float64(2)
memory usage: 612.0 bytes
```

In [12]:
```
df3.describe()
```

Out[12]:

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

In [13]:
```
df3.shape
```

Out[13]:
```
(30, 2)
```

In [14]:
```
df3.isnull().sum()
```

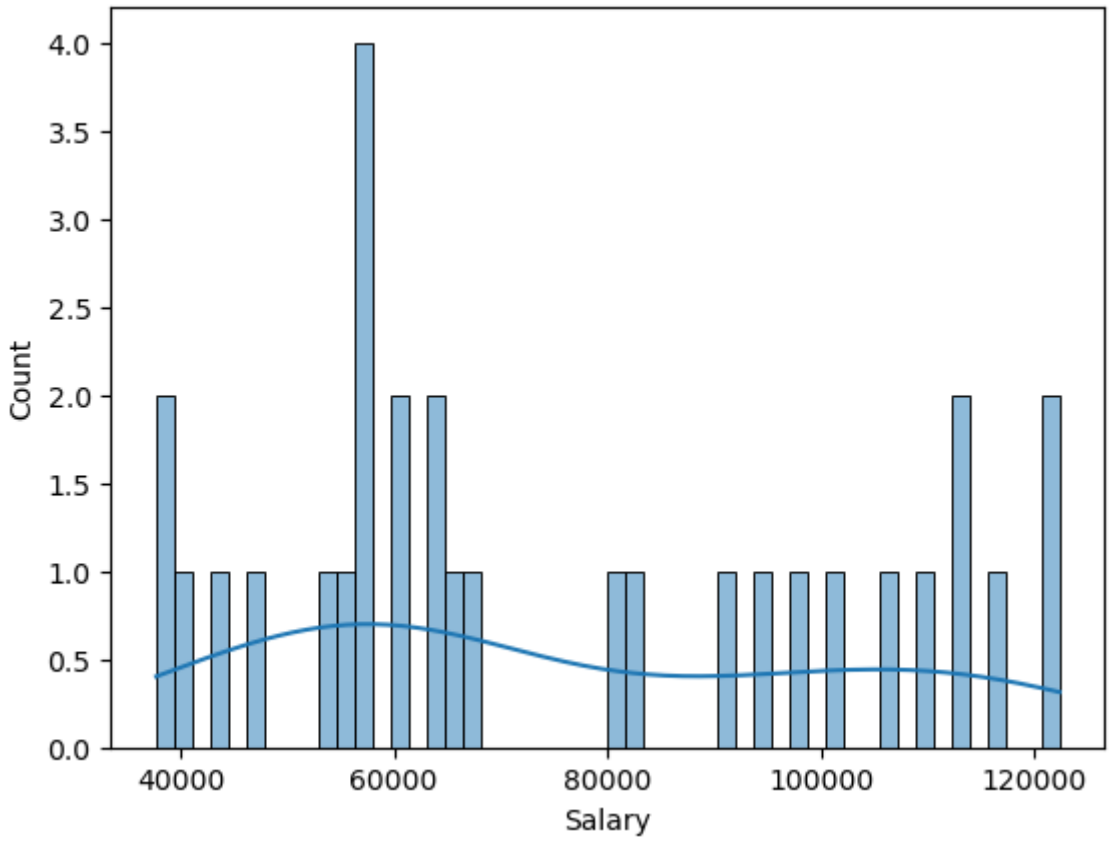
Out[14]:
```
YearsExperience    0
Salary             0
dtype: int64
```

In [15]:
```
df3.dtypes
```

Out[15]:
```
YearsExperience    float64
Salary             float64
dtype: object
```

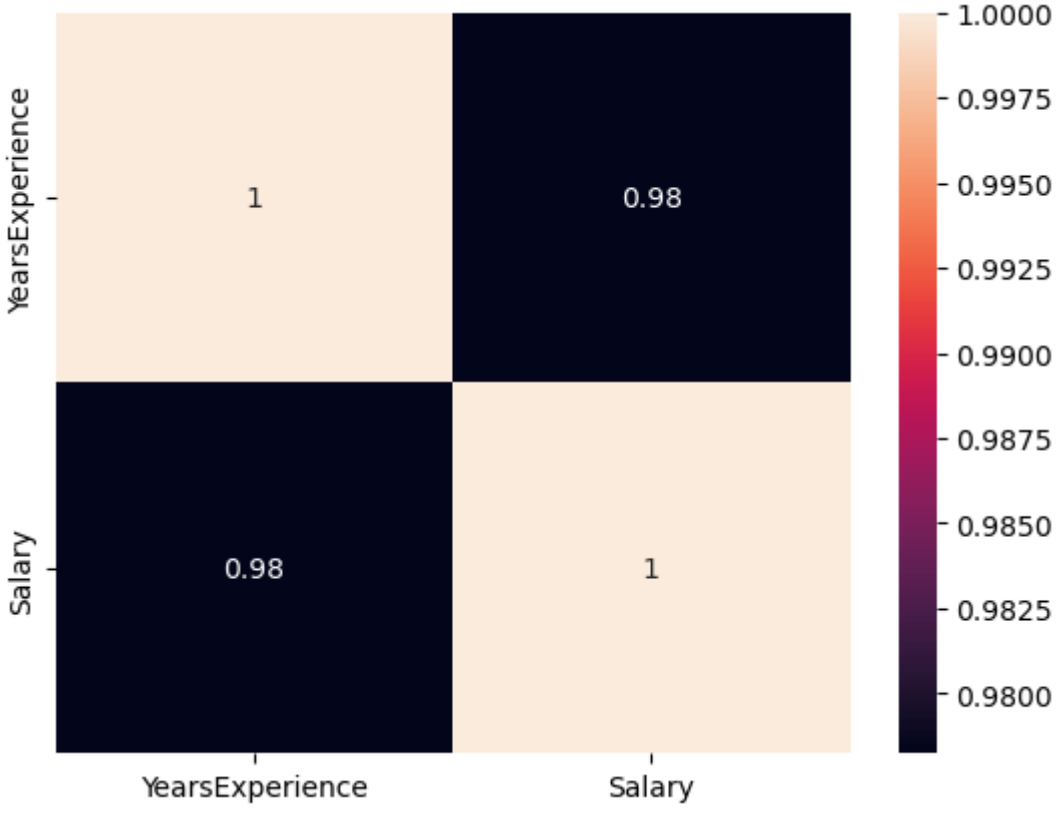
In [17]:
```
sns.histplot(data=df3, x='Salary', bins=50, kde=True)
```

Out[17]:
```
<Axes: xlabel='Salary', ylabel='Count'>
```



In [18]:
```
sns.heatmap(df3.corr(numeric_only = True),annot = True)
```

Out[18]: <Axes: >



In [20]:
```
df3.columns
```

Out[20]: Index(['YearsExperience', 'Salary'], dtype='object')

In [22]:
```
X = df3.drop('Salary', axis=1)
y = df3['Salary']
```

In [23]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error ,mean_squared_error,r2_sc
```

In [24]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.33,
```

In [25]:
```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)
```

Out[25]:

▾ LinearRegression

▶ Parameters

In [26]:
```
y_pred = model.predict(X_test)
y_pred
```

Out[26]:
```
array([115814.30756236,  71511.92534771, 102617.85328566,  75282.3408553
4,
        55487.65944028,  60200.67882482, 122412.53470072, 107330.8726702
,
        63028.49045554,  35692.97802523])
```

In [27]:
```
y_test
```

Out[27]:
```
27    112635.0
15     67938.0
23    113812.0
17     83088.0
8      64445.0
9      57189.0
28    122391.0
24    109431.0
12     56957.0
0      39343.0
Name: Salary, dtype: float64
```

In [29]:
```
r2 = r2_score(y_test,y_pred)
mse = mean_absolute_error(y_test,y_pred)
mae = mean_squared_error(y_test,y_pred)
print("R2 Score:",r2)
print("MSE:",mse)
print("MAE:",mae)
```

```
R2 Score: 0.9553063138044949
MSE: 4956.5232614449005
MAE: 35301898.887134895
```